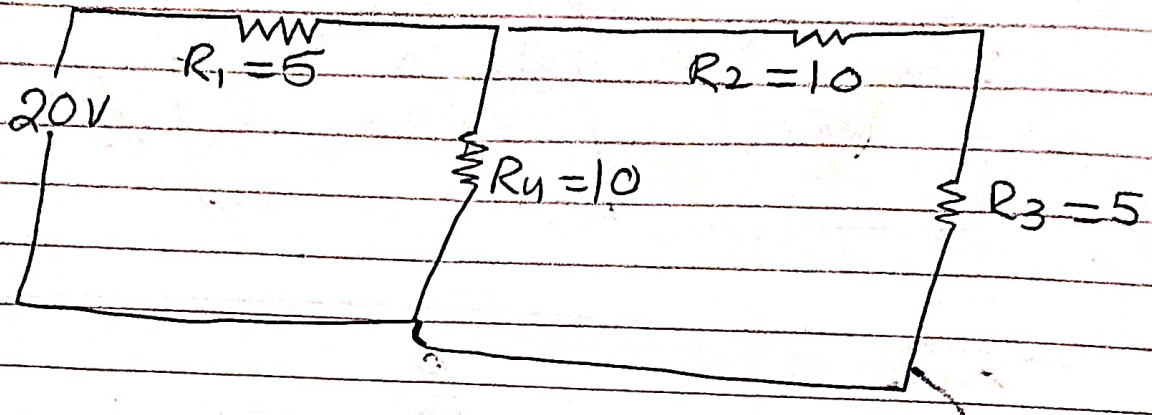


day/date

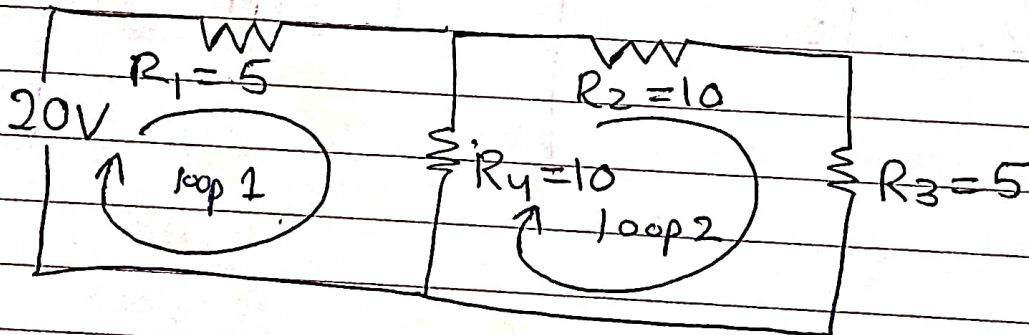
KVL

Example:



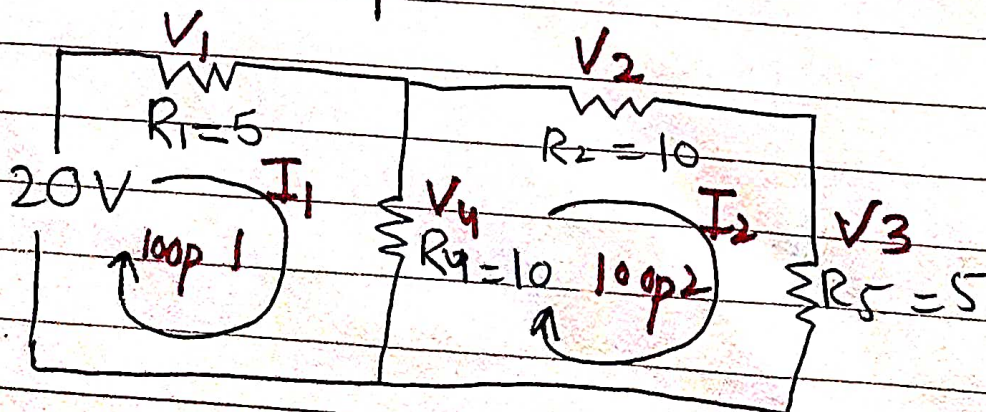
Step 1

Identify loops

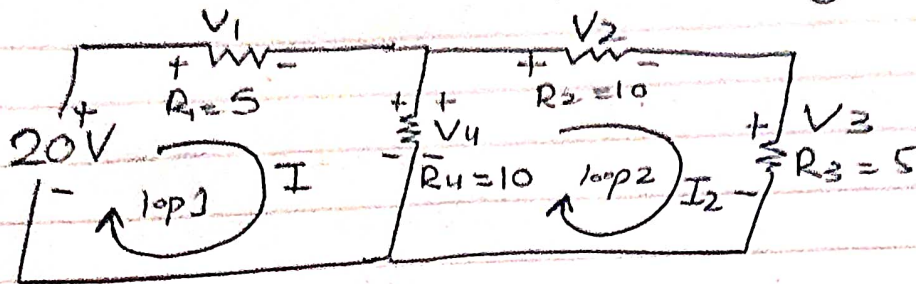


Step 2

the
Name loops :



Step 3 Mark polarity (+, -)



Step 4 Apply KVL on each loop

Loop 1

$$\begin{aligned}
 -20 + V_1 + V_4 &= 0 \\
 -20 + I_1 R_1 + (I_1 - I_2) R_4 &= 0 \\
 -20 + I_1 5 + (I_1 - I_2) 10 &= 0 \\
 -20 + 5I_1 + 10I_1 - 10I_2 &= 0 \\
 -20 + 15I_1 - 10I_2 &= 0 \\
 15I_1 - 10I_2 &= 20
 \end{aligned}$$

↓
eq ①

Loop 2

$$\begin{aligned}
 V_4 + V_2 + V_3 &= 0 \\
 (I_1 - I_2) R_4 + I_2 R_2 + I_2 R_3 &= 0 \\
 (I_1 - I_2) 10 + I_2 10 + I_2 5 &= 0 \\
 10I_1 - 10I_2 + 10I_2 + I_2 10 + I_2 5 &= 0 \\
 10I_1 + 25I_2 &= 0
 \end{aligned}$$

↓
eq 2

Now find eq 1:

$$15I_1 - 10I_2 = 20$$

~~$$15I_1 - 10I_2 = 20$$~~

$$I_1 = \frac{20 + 10I_2}{15}$$

↓
eq ③

day/date

Now put eq (3) into eq (2)

$$-10I_1 + 25I_2 = 0$$

$$-\overset{2}{\cancel{10}}(20 + \underset{\substack{15 \\ 3 \xrightarrow{\text{multiply}}}}{\cancel{10}}I_2) + 25I_2 = 0$$

$$40 + 20I_2 + 75I_2 = 0$$

$$I_2 = \frac{40}{55} = -0.73A$$

Now 0.73A put in eq (1)

$$15I_1 - 10I_2 = 20$$

$$15I_1 - 10(-0.73) = 20$$

~~$$15I_1 - 10(-0.73) = 20$$~~

~~$$15I_1 - 10(-0.73) = 20$$~~

~~$$15I_1$$~~

$$I_1 = 1.81 \quad , \quad I_2 = 0.72$$

Resistance	Current	voltage
$R_1 = 5$	1.81A	9.1V
$R_2 = 10$	0.72A	7.2V
$R_3 = 5$	0.72A	3.6V
$R_4 = 10$	1.81A	18.1V